Human Compression: the audit of a fixed corpus

Human Compression project

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1 The cutoff and the objects being compared

New manuscript intake is closed. The baseline is the Git commit

`3b067cc2eba4ac2f2f91248448727639d8ac7c5a`,

also named `intake-2026-10-11`. Its catalogue contains 161 original manuscripts. At that cutoff, 102 had retrieved full texts, 61 had specified encodings, 48 had a rederived result of some stated scope, and 27 have previously admitted recoveries of specified principal results. These overlapping counts do not describe 161 completed readings, 27 replacements of every assertion, or 27 fully formalized papers.

The programme covers every accessible original among these 161 manuscripts. Closing intake fixes the list of titles; it does not stop retrieval or proof work within that list. The 27 admitted recoveries are a progress count. The remaining accessible originals await their own complete proofs, reviews and measurements, with existing common theory reused wherever its hypotheses can be proved. F015, L028 and N007 remain in this programme.

An accessible original cannot be excluded merely because its proof is unfinished, its length is unmeasured, or it has not yet been assigned to a topic. A complete authored rendering is usable even without an original PDF, provided the representation is stated in the comparison. Only a genuinely unavailable source may be left out of the analysis after documented retrieval attempts; obtaining it later reopens the entry. A current numerical comparison may cover only completed principal-result contracts, but this does not remove the other accessible originals from the work.

The historical cutoff is recorded in `audit/intake.json`. The source register is `audit/source_registry.json`. Their earlier restrictions on carry-over work are superseded by this programme. The present inventory must record access, principal targets, proof status and measurement status separately for every original manuscript.

For each source, the audit first fixes an edition and a list of results to be recovered, with all parameter ranges and exceptional cases. A correction to a false source statement is recorded as a correction, not as a proof of the original statement. Recovery of the main results does not by itself recover every auxiliary assertion, table or illustration. The source and replacement must have the same declared mathematical outputs before their lengths are compared.

2 Theory, applications and proof closure

A topic A consists of a complete shared theory U_A and complete applications $A_1, \dots, A_k$, corresponding to declared source contracts $LA_1, \dots, LA_k$. Each application must check the actual hypotheses of the shared theorems and supply its local constructions, converse arguments, exceptions, geometric identifications and any necessary exhaustive computations. Similar wording is not a proof of common structure. A paragraph saying that a theorem applies is sufficient only when all its hypotheses and the claimed output have actually been identified.

Existing proofs supply material for new common arguments. The objective is to reprove the source results as consequences of a small theory, not to shorten their wording. Moving a proof to another file earns no mathematical credit. The earlier combined manuscripts are a proof bank, not the intended replacement. The current position is as follows.

Topic	Existing material and next obligation
A	Complete shared theory and six applications, with independent mathematical and source review and a fresh full program replay. Exact source corrections and retained inputs are explicit. The complete length comparison and paired dependency audit are independently accepted.
B	Complete common theory and four applications on forests and primitive coset geometries, independently reviewed for thirty declared clauses. The length and paired dependency audits are accepted; N001's PDF cost and the numerical whole-topic factor remain unknown.
C	Complete common theory and four applications, with an explicit binary translation inverse, full geometric identifications and the fresh Janko census. Complete lengths are accepted, with L108's PDF cost unknown. The paired dependency audit is accepted: exclusive passage costs conserve all complete documents and count shared prerequisites once.
D	Complete common theory and three applications, independently reviewed with all programs and ten scoped Lean declarations. Field and group classifications remain retained inputs; L104's principal contract excludes its printed diagrams, Aut/Cor groups and doubling relation. Complete lengths and all combined/conditional fractions are accepted. The paired dependency audit conserves every document total and charges each shared passage once.
E	A new common theorem for string markings with proper exceptional ends derives the intersection property instead of assuming it. Three applications recover the two polytope classifications and the corrected PSL maniplex results. Prior editions remain separate baselines.
F	Complete common orbit, character and cyclotomic theory and the L014 application, independently reviewed. All three programs are validation; five scoped Lean declarations verify subgroup interfaces and identities. The tightness converse retains its external classification theorem. Complete lengths and combined/conditional fractions are accepted; paired dependencies remain pending.
G	Complete flag, Steiner residue and Hermitian theory, with corrected L009 and L092 applications. All 79 source records, thirteen programs and forty scoped Lean declarations reviewed. External classifications and explicit exclusions remain. Complete lengths and combined/conditional fractions are accepted; paired dependencies remain pending.
Affine	Complete common affine theory and corrected L012 application, including all kernel, square and cubic arguments they use. All six full programs and nine scoped Lean declarations passed independent review. Complete lengths and combined/conditional fractions are accepted; the paired dependency audit conserves every complete document and charges shared passages once.

Finite pairs	Complete recognition, chamber and cleaning theory, with the F010 application and both complete validation programs. All six source blocks and three qualifications were independently reviewed. Twelve scoped Lean declarations support the written proofs; lengths and paired dependencies are separate. The length audit is accepted, with the small written saving reversing under UTF-8 counting; paired dependencies remain pending.
Chamber	Complete common theory and L008 and N016 applications, with the explicit reflection, every required family and all six finite groups. All 27 scope rows and fourteen scoped Lean declarations independently reviewed. Source qualifications and retained geometric/classification inputs remain explicit. Complete lengths and combined/conditional fractions are accepted; paired dependencies remain pending.

The first publication migration supplies ten editable TeX proof texts: two in A, three in E, two in G, and three separate cover results. It preserved their PDF text and page counts. Later editions supply the complete theory and applications listed above. Each topic requires its own source, proof, length and dependency audits; preserving a text does not itself establish new mathematics.

3 Attribution and formal verification

Each application will have a machine-readable ledger connecting a source result and its exact locator to the replacement theorem, its proof, retained inputs, required programs, and any Lean declarations. The TeX edition will distinguish borrowed results, contributions of this project, recovery of several previous results, source corrections, computational checks, formalized lemmas, and answers to previously posed questions. An assertion of historical novelty requires a separate literature audit.

For every formal claim, the audit inspects the actual Lean statement, including its hypotheses. A general transport lemma with an unproved geometric hypothesis does not formalize the geometric theorem. The ledger must name the written obligations left outside Lean. Finite computations count as formal evidence only to the extent that their certificates and the implication to the claimed theorem are checked.

The canonical library now contains 100 modules. Its build evidence combines one fresh run of 96 unchanged modules with four separately checked additions: the marked-extension module, two normal-kernel modules and the commuting retraction module. The latter has four checked definitions and seven checked theorems. No single fresh run of all 100 modules is asserted. All belong in the default build. Source hashes, toolchain, dependency revisions and axiom reports are recorded. One unchanged legacy module has four native-evaluation axioms. The generic compatible-pair inverse is formalized; the application coordinate maps and their joint faithfulness remain written obligations. Existing limited formalizations will be extended where they verify substantive steps in our own proofs. Neither successful compilation nor the number of Lean declarations measures the fraction of a paper that has been formalized.

Topic B supplies a separate freshly checked Lean module, outside that 100-module default library. Its six declarations prove the actual residual-generation induction and its subgroup consequences. The geometric adapter, weighted-forest theorem and four complete applications remain written mathematics. The public packet includes the precise statement map, standard-axiom reports and finite witnesses to the necessity of its hypotheses. No source contract has acquired an end-to-end formalization.

Topic C has a separate three-module check: nine new declarations for the explicit semidirect inverse, plus seven unchanged displacement and intersection declarations. All sixteen were

freshly compiled by the independent reviewer and by root. The augmentation basis, source-specific maps and geometry, and the Janko census remain outside Lean. Its full finite census was rerun independently, using both complete required programs and the retained subgroup catalogue. The legacy-plus-B runner does not yet dispatch Topic C; its separate command is published with the edition.

The affine edition has nine separately replayed declarations: four reused quotient-intersection criteria, three reused sign-obstruction lemmas and two statements preserving intersections under smaller normal periods. Actual compiled types and hypotheses were inspected. Affine coordinates, presentation rewriting, catalogue completeness and the full geometric classification remain outside these Lean proofs. All six complete mathematical programs were independently replayed, including the full order-384 nonexistence search. This does not add an end-to-end source formalization or a new solved-question credit.

The finite-pair edition adds eight generic Lean declarations for actual restriction, rigidity, extension and subgroup transport. Four reused local declarations verify hexagon reflection facts. All twelve were independently compiled from source with standard axioms. The concrete graph hypotheses, component argument, chamber and cleaning constructions, recursive family and complete representation theorem remain written proofs. Both full programs validate bounded cases; neither is a uniform proof premise.

The chamber edition adds seven declarations for the actual reflection on two copies of a group, including adjacency and translation compatibility. Seven reused declarations remain compiled, with used and unused statements identified. Geometry reconstruction, retained classifications and the six exact GAP groups remain outside Lean. All 27 admitted principal contracts now have complete canonical common theory and applications; this does not complete their length, dependency or formalization audits.

4 Length and reuse

Fix the source contracts, representations, document boundaries and length convention c before reporting a factor. The intended principal comparison uses the full source manuscripts once their declared main-result contracts have been recovered. Ancillary material need not all be reproduced, but the omitted results and any corrections must remain explicit. A comparison restricted to matched proof passages is a separate sensitivity measure; it must not silently replace the full-manuscript numerator. The two headline measures are a complete topic and each conditional result. Write

$$R_A^{\mathrm{written}} = \frac{\sum_{i=1}^{k} c(LA_i)}{c(U_A) + \sum_{i=1}^{k} c(A_i)}, \qquad r_{A,i} \mid (U_A, P_A) = \frac{c(LA_i)}{c(A_i)}.$$

The second measure assumes that the *whole* umbrella and all required programs P_A have already been acquired, including any application-specific parts of those programs. Their acquisition costs must be displayed. Beside the written-topic factor, report the complete required-program coordinate

$$R_A^{\mathrm{required}} = \frac{\sum_{i=1}^{k} c(LA_i)}{c(U_A) + \sum_{i=1}^{k} c(A_i) + c(P_A)}.$$

Here P_A contains required proof artifacts not already counted in the written papers. It is charged once within the topic, without hypothetical pruning. A printed program appendix and its raw source are alternative representations, not two charges for the same program. Written and program costs use compatible units: pages are never added to code bytes. Optional diagnostic programs have a separate coordinate. If any required artifact is excluded from the prepaid library in a different comparison, its actual cost must be added to the relevant application's denominator. Classical inputs and unavailable historical artifact costs remain explicit.

The complete cost of U_B is charged in topic B, even if it overlaps U_A. Conversely, U_A is not charged again to every application in topic A. Conditional application factors must display the cost of the assumed umbrella. Missing costs are unknown, not zero. The current three-application Topic E edition costs 39,303 non-whitespace characters against 71,375 in its three sources. Its complete written factor is 1.816×, or 1.523× including the complete required program. Given the whole common theory and program, the application factors are 6.957×, 11.271× and 2.018×. The earlier two-source and three-source editions remain separately bound baselines; their costs are not substituted for those of the current proofs.

Topic A now has six applications under the same rule. Its six sources cost 195,379 characters, against 49,059 for the full common theory and 78,251 for the applications. The written factor is 1.535×; the complete program adds 53,035, giving 1.083×. Given the whole common theory and program, the six factors, in the declared order, are 2.016×, 3.706×, 2.853×, 4.107×, 3.519× and 1.310×. The source-image and retained-input limitations below apply to these figures.

Topic B charges 14,119 characters for its common theory and 44,942 for all four applications, giving a complete written denominator of 59,061. The three available source PDFs cost 57,539. If the unavailable N001 PDF has cost x, the complete comparison is $(x + 57{,}539)/59{,}061$; there is no numerical overall factor. B1 remains charged. After the whole common theory has been acquired, the factors for N001, N003, N008 and L030 are unknown, 1.254×, 1.576× and 1.811×, respectively. No mathematical enumeration program is required by this proof edition.

Topic D's three sources cost 47,894 characters. Its whole common paper costs 15,632 and all three applications cost 29,855. The complete written factor is therefore $47{,}894/45{,}487 = 1.053\times$. The whole printed program appendix adds 27,070, giving $47{,}894/72{,}557 = 0.660\times$; the program cost outweighs the written saving. Using all eight complete raw programs instead gives 0.668×, an alternative representation, not an extra charge. After acquiring the whole theory and all eight programs, including the three full diagnostics, the conditional factors are 1.903×, 1.546× and 1.149×. The two complete execution drivers are separately inventoried at 10,038 characters. Historical software, external proofs, catalogue construction and graphics have unknown costs. These are comparisons of the declared available texts, not estimates of total historical effort or minimum proof complexity.

The affine topic compares one complete source, costing 32,767 characters, with a whole common paper costing 19,070 and an application costing 22,638. The complete written replacement is longer, with factor $32{,}767/41{,}708 = 0.786\times$. After acquiring the whole common paper and all six required programs, the local factor is $32{,}767/22{,}638 = 1.447\times$. The full printed program appendix adds 17,250, giving 0.556×; all six raw files give 0.561× as an alternative representation. Three complete auxiliary scripts cost 16,581 and three formal files cost 7,958 separately, outside those denominators. The principal contract, source corrections and exclusions are unchanged. Its accepted paired dependency audit conserves all full-document costs and charges shared passages once in each prerequisite union.

Topic F's complete source costs 25,115 characters, its whole common theory 7,118 and its application 12,744: the combined factor is $25{,}115/19{,}862 = 1.264\times$. After acquiring the whole theory and validation package, the individual factor is $25{,}115/12{,}744 = 1.971\times$. Its three programs validate examples; the uniform proofs do not depend on them. Their complete appendix adds 8,837, giving 0.875×; the raw files cost 8,061 instead, giving 0.899×. Two drivers cost 10,881 and the Lean file 2,269 separately. Tightness classification and graphics retain unknown costs. The corrected principal contract and exclusions apply to both factors; paired dependencies remain pending.

Topic G's two complete sources cost 73,199 characters. Its whole common paper costs 30,928 and its applications 23,544 and 61,663, giving the combined written factor $73{,}199/116{,}135 = 0.630\times$. The current complete replacement is longer. After acquiring the whole common paper and all thirteen complete programs, the individual factors are $49{,}891/23{,}544 = 2.119\times$ and $23{,}308/61{,}663 = 0.378\times$; the second application remains longer even then. The complete printed

program appendix adds 53,018, giving 0.433×. All thirteen raw programs instead add 50,184, giving 0.440×, as an alternative representation. Four finite premises, two witness or certificate bridges and seven validations remain distinguished. Three drivers cost 14,919 and six Lean files 19,480 separately. Retained classification proofs, historical programs and graphics have unknown full costs. The publisher cover remains in the primary source numerator; its removal is a separate sensitivity. Source corrections and exclusions remain explicit, and paired dependency costs remain under review.

The finite-pair representation topic compares a source costing 25,348 characters with common theory costing 11,288 and an application costing 13,688. Its combined written factor is $25{,}348/24{,}976 = 1.015\times$, a difference of only 372 characters. After acquiring the whole theory and both complete validation programs, the individual factor is $25{,}348/13{,}688 = 1.852\times$. The full printed appendix adds 9,025, giving 0.746×; the two complete raw programs instead add 8,310, giving 0.762×. Neither program is a premise of the uniform proof. The two historical drivers cost 13,275 and the two Lean files 5,487 separately. Under calibrated NFC UTF-8 bytes, the written factor is $31{,}880/32{,}836 = 0.971\times$: the small saving is not consistent across units. Original attribution, source qualifications and unknown external and graphical costs remain explicit. The source's Theorem A already answers its Question 2.4; no new answer is credited.

The primary text coordinate is extracted PDF text, normalized under a fixed protocol. Exact running page furniture is excluded, while titles, abstracts, references and other document text remain charged. Both NFC and NFKC normalizations are retained, with five length conventions each: whitespace words, alphanumeric units, non-whitespace characters, characters with collapsed whitespace, and UTF-8 bytes with collapsed whitespace. Necessary font-to-Unicode calibrations are bound to actual glyph occurrences and reviewed independently. Raw extraction is retained as a sensitivity.

Some scientific diagrams and labels are images. Their absence from extraction does not make their cost zero. In particular, the L035 source in Topic A has scientific image content, including a nonblank page whose extracted text is empty. Its text factor therefore measures complete-document extractable text, not all visible content. Page counts and known image limitations are reported separately. Missing graphic, historical-program and external-proof costs remain unknown; program-inclusive factors are comparisons of known costs.

A separate sequential account follows the reuse discipline employed in the GKM project. Let K_{i-1} be the already paid-for library and D_i the full dependency closure of the next application, in an additive representation of identified proof units. Then

$$\Delta_i = c(D_i \setminus K_{i-1}), \qquad K_i = K_{i-1} \cup D_i.$$

Hypothesis checks, new interfaces and required programs belong to D_i. The fixed graph and an explicit admissible order determine this account. Without unrecorded revisions, the increments telescope to the cost of the acquired union; no earlier theory disappears from the total. This sequential account does not alter the independent topic costs.

Both the source and replacement require theorem-dependency graphs with the same output contracts. Complete proof units, including exposition and interfaces, receive costs. A citation alone does not establish that a result is used; a cited theorem that is actually used remains an explicit retained-input edge. Publish per-paper costs, cumulative costs and sequential acquisition costs. A sawtooth pattern must emerge from those measurements. It is neither an assumption of the audit nor an established conclusion of the current work.

5 Dependency nodes and their costs

For each matched result, construct a directed acyclic graph whose arrows run from a prerequisite to the argument using it. Numbered theorems, lemmas, propositions and corollaries give a first level of proof units. Substantive unnumbered arguments receive explicit units too; definitions,

context, references and programs are distinguished from mathematical proof nodes. A proof node contains its full statement, proof and local hypothesis checks. Repeating a statement does not create a second result, but the printed repetition still has a cost.

For an output t, let $D(t)$ contain t and all its prerequisites, and let $P(v)$ be the set of passage owners used by node v. We report the number $N(t)$ of mathematical proof nodes, the external inputs, and

$$\mathcal{P}(t) = \bigcup_{v \in D(t)} P(v), \qquad W_{\text{known}}(t) = \sum_{p \in \mathcal{P}(t),\, c(p) \text{ known}} c(p).$$

Several nodes may refer to the same statement or setup passage. Taking the union before summation charges that passage once. Adding the displayed node costs would in general count some passages twice. The omitted unknown costs are listed explicitly; this sum is never presented as an unconditional total. Statements retained from a classical library have their printed statement cost and a separate unknown proof cost. Required programs are charged with the proof they support.

Each measured document is partitioned into disjoint text spans. A span has exactly one cost owner. A logical node refers to all owners needed for its complete statement, proof and local checks; these references may overlap with another node's. Context outside the logical closure remains charged in the full-manuscript comparison. This makes the weighted graph additive without replacing the primary full-document numerator by a selected-proof numerator. Edges record actual use with a locator, not merely the appearance of a citation.

Node counts depend on granularity. A theorem containing six cases must not earn a sixfold saving simply because another author numbered the cases as lemmas. We therefore retain the named-result count and expose substantive branches where bundling matters. Parent and child descriptions are two views of the same proof: costs belong to the leaves and are not charged again at their parents. The weighted total must be invariant under such regrouping. Counts, weighted costs and their granularity are reported together; no raw node ratio is called a complexity ratio.

For several outputs the acquired closure is their union, so a common proof is charged once within the topic. After acquiring U_A, the next application pays its new closure and its own hypothesis checks. Source papers receive the same reuse treatment for earlier results they actually invoke. Separate topics still pay their whole umbrellas independently, as in the preceding section. The paired graphs should identify where a new argument eliminates a chain of old arguments, where it merely moves one, and where an unchanged external theorem is still required.

6 Questions and publication

Answers to open questions are assessed separately from recovery of known results. Each entry states the exact question and edition, the answer's scope, proof dependencies, computational and formal evidence, and limits of the priority search. They are not inserted into a historical-length numerator. In particular an absent historical proof has no measured length, so an infinite compression factor is not an empirical length ratio. The project may identify a qualitative gain beyond recovery while keeping this numerical distinction explicit.

TeX and Lean are the editable mathematical sources; PDFs are readers and JSON supplies ledgers. Markdown stays in local working directories. The 992 historical Markdown files remain bound to their exact Git objects, with their paths and hashes in `audit/archive_markdown.json`. The current verifier ignores local notes and checks the archived bytes separately from the current tracked publication. The old replay programs run from a complete checkout of the pinned intake commit. They are not represented as fresh replays of the reorganized publication.

The matched dependency audits for Topics E, A, B, C, D and the affine topic are complete within their stated scope. Topic A assigns all 341 counted text spans to 337 exclusive owners; its

236 argument and reference nodes and 392 edges do not supply a ratio of node counts. All 140 document-cost coordinates agree with the complete-manuscript measurement. Topic B assigns 217 text spans to 214 exclusive owners, with 184 argument and reference nodes and 348 edges. Its eighty document-cost coordinates also agree with the complete-manuscript measurement. Topic C has 156 nodes, 249 typed arrows and 199 passage owners; all 270 document metric cells are conserved, while L108's source cost remains unknown. Topic D has 137 nodes, 284 typed arrows and 220 passage owners. Its 240 document metric cells and 116 prerequisite unions preserve the complete written and program comparisons. The current D graph records one corrected premise and two restored shared setup passages. None of these graphs supplies a compression factor from its number of nodes. The affine graph has 119 nodes, 228 typed arrows and 194 exclusive owners. All 120 document cells and 222 prerequisite unions are conserved. Its corrected sentence boundary changes local passage costs without changing any complete comparison. The remaining audits use existing proofs. Acceptance requires independent mathematical review, complete attribution, the declared Lean and computation checks, and all required costs in the denominator. Closing intake adds no mathematical credit.